

Partial Differential Equations

Problem Sheet 2

1. If u solves $u_x + u_y = 0$, then u is constant along the lines parallel to $y = x$.
2. Find the solution of the linear equation $u_x + xu_y = y$ satisfying the initial condition

$$u(0, y) = g(y), \quad y \in \mathbb{R}.$$

Also, draw the projected characteristics intersecting the given initial curve.

3. Solve

$$xu_x + (x + y)u_y = 1,$$

subject to the initial condition

$$u(1, y) = y, \quad -\infty < y < \infty.$$

Is the solution defined everywhere?

4. Does there exist a solution for the following problem? If not, justify your answer. If yes, find the solution:

$$\begin{cases} yu_x - xu_y = xy & \text{in } \mathbb{R}^2, \\ u(x, 0) = x^2, & x \in \mathbb{R}. \end{cases}$$

5. Solve the following Cauchy problem:

$$\begin{cases} u_x + yu_y = 1 & \text{in } \mathbb{R}^2, \\ u(0, y) = f(y), & y \in \mathbb{R}, \end{cases} \quad f \in C^1(\mathbb{R}),$$

and determine whether the solution is unique. Also, determine the domain of existence of the solution.

6. Solve

$$uu_x + u_y = 1,$$

subject to the initial condition

$$u(3x, 2) = 4 - 3x, \quad -\infty < x < \infty.$$

7. Consider the PDE

$$xu_x + yu_y = 2u, \quad -\infty < x < \infty, \quad -\infty < y < \infty.$$

- (a) Show that $u(x, y) = x^2 + y^2$ is a solution when $u(x, 0) = x^2$.
- (b) Solve using method of characteristics and show that $u(x, y) = x^2$ is a solution when the initial condition is $u(x, 0) = x^2$.
- (c) Why do you think the uniqueness is lost?